

Red Blood Cell Transfusion Threshold in Postsurgical Pediatric Intensive Care Patients

A Randomized Clinical Trial

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Background: The optimal transfusion threshold after surgery in children is unknown. We analyzed the general surgery subgroup of the TRIPICU (Transfusion Requirements in Pediatric Intensive Care Units) study to determine the impact of a restrictive versus a liberal transfusion strategy on new or progressive multiple organ dysfunction syndrome (MODS).

Methods: The TRIPICU study, a prospective randomized controlled trial conducted in 17 centers, enrolled a total of 648 critically ill children with a hemoglobin equal to or below 9.5 g/dL within 7 days of pediatric intensive care unit (PICU) admission to receive prestorage leukocyte-reduced red-cell transfusion if their hemoglobin dropped below either 7.0 g/dL (restrictive) or 9.5 g/dL (liberal). A subgroup of 124 postoperative patients (60 randomized to restrictive and 64 to the liberal group) were analyzed. This study was registered at <http://www.controlled-trials.com> and carries the following ID ISRCTN37246456.

Results: Participants in the restrictive and liberal groups were similar at randomization in age (restrictive vs. liberal: 53.5 ± 51.8 vs. 73.7 ± 61.8 months), severity of illness (pediatric risk of mortality [PRISM] score: 3.5 ± 4.0 vs. 4.4 ± 4.0), MODS (35% vs. 29%), need for mechanical ventilation (77% vs. 74%), and hemoglobin level (7.7 ± 1.1 vs. 7.9 ± 1.0 g/dL). The mean hemoglobin level remained 2.3 g/dL lower in the restrictive group after randomization. No significant differences were found for new or progressive MODS (8% vs. 9%; $P = 0.83$) or for 28-day mortality (2% vs. 2%; $P = 0.96$) in the restrictive versus liberal group. However, there was a statistically significant difference between groups for PICU length of stay (7.7 ± 6.6 days for the restrictive group vs. 11.6 ± 10.2 days for the liberal group; $P = 0.03$).

Conclusions: In this subgroup analysis of pediatric general surgery patients, we found no conclusive evidence that a restrictive red-cell transfusion strategy, as compared with a liberal one, increased the rate of new or progressive MODS or mortality.

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In recent years, red blood cell transfusion strategies and triggers have stimulated great interest, leading to much debate regarding the benefits of aggressive transfusion strategies and indications for transfusing blood products. There have been several studies on the appropriate transfusion threshold for critically ill patients.^{1–5} However there is a paucity of data on optimal transfusion strategies

following surgery, particularly concerning the critically ill pediatric surgical patient. The TRIPICU study was a randomized controlled trial (RCT) that was conducted among pediatric intensive care unit (PICU) patients. It demonstrated that a restrictive strategy in this population was not associated with a different outcome compared with the liberal strategy.⁶

Because of the lack of clear guidelines concerning transfusion threshold for PICU surgical patients, and because of the high variability in transfusion practice for those patients,^{7,8} a subgroup analysis dedicated to surgical patients was designed in the original TRIPICU protocol. Thus, in the present subgroup analysis, we compare the effect of restrictive and liberal transfusion strategies on multiple organ dysfunction and adverse outcomes in pediatric surgical patients admitted to the PICU.

METHODS

The TRIPICU Study

Protocol

A detailed description of the TRIPICU results was reported previously.⁶ Briefly, the TRIPICU study enrolled stabilized critically ill children from 19 tertiary-care PICUs. The study protocol was approved by institutional review boards, and parental consent was obtained. The condition of patients was considered stable if the mean systemic arterial pressure was not less than 2 SD below the normal mean for age and if cardiovascular treatment (catecholamines and volume administered) had not been increased for at least 2 hours before enrollment. Once stabilized, children aged between 3 days and 14 years, with at least one hemoglobin concentration ≤ 9.5 g/dL within the first 7 days after PICU admission were considered for inclusion. TRIPICU study exclusion criteria are listed in Figure 1.

Participants were randomly allocated to restrictive or liberal treatment arms. In the restrictive group, the transfusion threshold was hemoglobin of 7.0 g/dL with a target range after the transfusion between 8.5 and 9.5 g/dL; in the liberal group, the threshold was 9.5 g/dL with a target range of 11.0 to 12.0 g/dL. Only prestorage leukocyte-reduced allogeneic red cell units were used. Transfusion strategies were applied for up to 28 days postrandomization or until the time of death, whichever came first. Temporary suspensions of the protocol were allowed during active blood loss, surgery done on an emergency basis, severe hypoxemia, or hemodynamic instability.

The primary outcome was the proportion of patients who developed or had progression of multiple organ dysfunction syndrome (MODS), as defined by Proulx et al.⁹ We also looked at markers of severity of MODS (the highest number of organ dysfunction per patient and the Pediatric Logistic Organ Dysfunction [PELOD] score),¹⁰ and we collected information on secondary outcomes including nosocomial infections, mortality, duration of mechanical ventilation and PICU length of stay.¹¹

Continuous variables were compared using Student *t* test or Wilcoxon Rank Sum test; categorical variables were analyzed using

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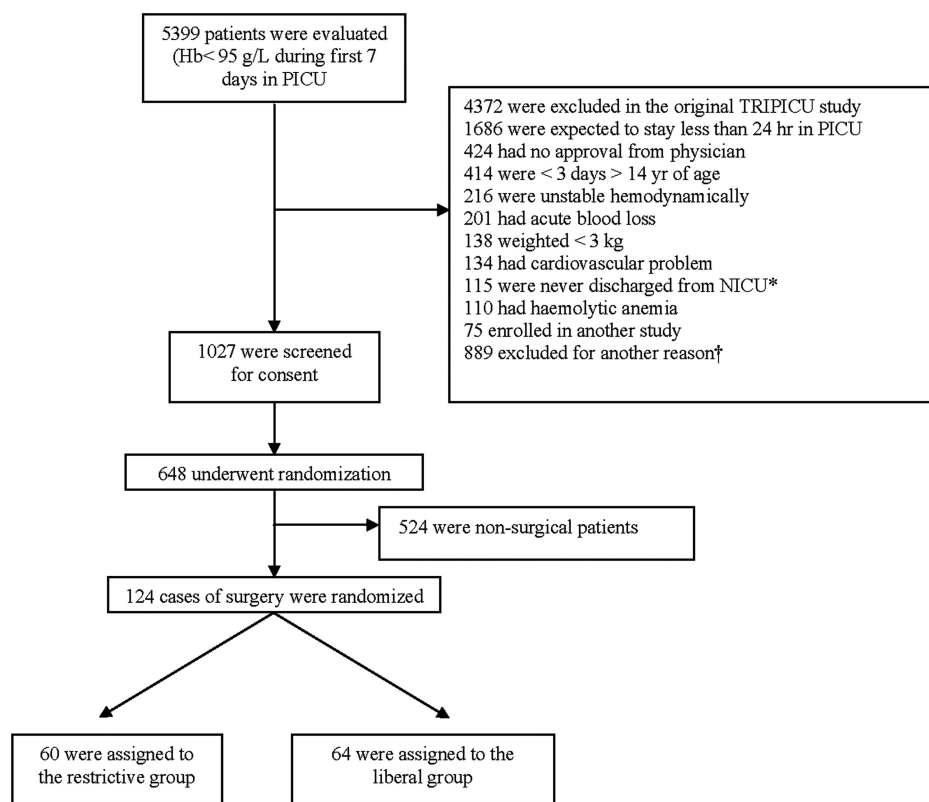
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FIGURE 1. *NICU: Neonatal Intensive Care Unit.†In addition to the causes listed for exclusion, other causes were a postconception age of less than 40 weeks (69 patients), severe thrombocytopenia (68), hypoxemia (65), a decision to withhold or withdraw critical care (59), predicted survival of less than 24 hours (54), previous enrolment in the study (33), brain death (25), extracorporeal membrane oxygenation (22), hemofiltration (21), blood exchange transfusion (20), plasmapheresis (17), an inability to receive blood products (14), pregnancy (1), and 391 for miscellaneous reasons. Some patients in PICUs had more than one exclusion criterion.



χ^2 testing. Baseline characteristics were compared using univariate descriptive statistics followed by logistic regression to evaluate the effect on outcomes of clinically important covariates including patient age, country, and PRISM scores.¹² Descriptive statistics are reported as mean $\pm$ SD.

It was estimated that at least 626 patients would be required to complete the TRIPICU study and to test a noninferiority hypothesis ($P < 0.05$; power of 0.9; margin of safety: absolute risk reduction of 10%). This subgroup analysis involved 124 surgical patients and the statistical analyses tested noninferiority. Statistical analysis of the primary outcome measure was conducted using an “intent-to-treat” approach. In the primary analysis, we calculated 95% confidence intervals (95% CI) around the absolute risk reduction in the proportion of patients with new or progressive MODS. We also conducted a per-protocol analysis of the primary outcome in patients who met or exceeded an 80% rate of protocol adherence. Adherence was defined as the proportion of days after randomization during which at least one hemoglobin concentration was above the transfusion threshold.

All secondary analyses were conducted using an “intent-to-treat” approach. We compared daily PELOD scores, using the worst scores after baseline, and the total number of organ dysfunctions per patient. We also compared 28-day and hospital all-cause mortality, nosocomial infections, sepsis, transfusion reactions, duration of mechanical ventilation, and PICU and hospital length of stay. To determine if a restrictive transfusion strategy decreased exposure to red cells, we compared the total number of transfusions per patient and the proportion of patients who did not receive red cell transfusions in the 2 groups.

Differences were considered statistically significant when a 2-sided alpha level was <0.05 . No adjustments were made for multiple comparisons. Data were analyzed by a biostatistician (TD) with SAS software (version 9.1, SAS Institute).

Assignment

Randomization was centralized, with assignment data posted on the internet. Patients were assigned to the study groups in blocks of 2 or 4 that were randomly distributed and stratified according to center and 3 age groups (≤ 28 days, 29–364 days, and >364 days). Physicians, nurses, and research staff were unaware of the block randomization strategy.

Blinding Procedures

The TRIPICU study was not blinded because this was not feasible. Clinical staff and parents were aware of the assignments to study groups (presence of blood bag at the bedside, increase in hemoglobin level post-transfusion), but the statistician and members of the data and safety monitoring committee were unaware of the assignments.

The Subgroup Analysis: Patients and Sites

The present study is a subgroup analysis of general surgery patients enrolled in the TRIPICU study. This subgroup analysis was planned prior to unblinding of data. Expected direction of results was stated before analysis was begun. Nonsurgical patients and patients who underwent any form of cardiac surgery were excluded from this subgroup. The baseline diagnosis and the types of surgery performed were recorded for the included patients.

RESULTS

Patients and Treatment Assignment

There were 124 general surgical pediatric patients enrolled from 17 sites and 4 countries in the general surgery subgroup, representing 19.5% of all the TRIPICU patients, of which 60 were randomized to the restrictive and 64 to the liberal group (Fig. 1). Table 1 displays the patient characteristics on admission to PICU.

TABLE 1. Characteristics of Patients at PICU* Entry†

	Restrictive	Liberal
No. patients per group	60	64
Data at entry in PICU		
Age (mo)	53.5 ± 51.8	73.7 ± 61.8
Weight (kg)	17.3 ± 12.9	23.4 ± 17.9
Gender (male) (%)	37 (62)	35 (54)
Severity of illness (PRISM score)‡	7.2 ± 4.8	7.2 ± 5.4
Mechanically ventilated (%)	46 (77)	48 (75)
Surgeries (no. patients [%])		
Chest	6 (9)	7 (9)
Transplantation	3 (5)	3 (4)
Neurosurgery	9 (14)	11 (15)
Scoliosis	6 (9)	9 (12)
Abdominal	21 (32)	20 (27)
Otorhinolaryngology§	6 (9)	8 (11)
Plastic surgery¶	5 (8)	7 (9)
Orthopedic surgery nonscoliosis	7 (11)	8 (11)
Others**	2 (3)	2 (3)
Total number of surgeries††	65	75
Red-cell transfusion between PICU admission and randomization		
No. transfused patients (%)	8 (13)	19 (30)
Red-cells (mL/kg) per transfused patient	10.8 ± 3.7	12.2 ± 3.5
Red-cell units (number) per transfused patient	0.2 ± 0.8	0.4 ± 0.9
Length of storage (d)	17.9 ± 9.5	14.1 ± 9.8

*PICU: pediatric intensive care unit.

†Plus-minus values are means ± SD.

‡Pediatric Risk of Mortality (PRISM) score ranges from 0 to 76, with higher scores indicating a greater risk of death.

§Laryngo/tracheoplasty, neck dissection and drainage, ORL fracture repair, urgent tracheotomy, cricoid split and tracheal graft, laryngopharyngeal cleft repair.

¶Cranial vault reconstruction, craniofacial surgery, facial reconstruction, skin graft, surgery for facial bones fractures, fasciotomy.

||Pelvic surgery, posterior spinal instrumentation for Chance fracture, femur fracture and stabilization, fixation of hip and tibia and fibula, orthopedic C2 fixation.

**Rigid bronchoscopy, thoracocentesis for decortication, bronchoscopy.

††The number of surgeries exceeds the number of patients because some patients underwent more than one surgery; percentage may not sum to 100 because of rounding.

TABLE 2. Data at Randomization*

	Restrictive	Liberal
No. patients per group	60	64
Hemoglobin level (g/dL)	7.7 ± 1.1	7.9 ± 1.0
Length of stay in PICU (days)	1.8 ± 1.6	2.2 ± 1.7
Age (3 strata)		
≤28 d	2 (3)	0 (0)
29–364 d	12 (20)	14 (22)
>365 d	46 (77)	50 (78)
Sites		
Belgium (3 sites) (%)	12 (20)	14 (22)
Canada (10 sites) (%)	37 (62)	40 (63)
United Kingdom (2 sites) (%)	4 (7)	3 (5)
United States (2 sites) (%)	7 (12)	7 (11)
Mechanically ventilated (%)	46 (77)	48 (74)
Severity of illness (PRISM score)†	3.5 ± 4.0	4.4 ± 4.0
Septic states‡		
Systemic inflammatory response syndrome (%)	36 (60)	35 (55)
Sepsis (%)	12 (20)	10 (16)
Severe sepsis (%)	5 (8)	7 (11)
Septic shock (%)		
Multiple organ dysfunction syndrome (MODS) (%)§	21 (35)	18 (29)
Respiratory dysfunction (%)	39 (65)	45 (70)
Cardiovascular dysfunction (%)	4 (7)	10 (16)
Hematological dysfunction	13 (22)	10 (16)
Neurological dysfunction (%)	2 (3)	2 (3)
Gastrointestinal or hepatic dysfunction (%)	1 (2)	3 (5)
Renal dysfunction (%)	0 (0)	3 (5)
Daily PELOD score on day 1§	5.3 ± 6.3	4.9 ± 5.4
Organ dysfunctions (number) per patient	1.2 ± 0.8	1.2 ± 1.0
No. patients with vasoactive drugs (%)¶	19 (32)	24 (38)
Corticosteroids (%)	12 (20)	11 (17)

*Plus-minus values are means ± SD.

†Pediatric Risk of Mortality (PRISM) score ranges from 0 to 76, with higher scores indicating a greater risk of death.

‡Organ dysfunction was determined as defined by Proulx et al.⁹

§Pediatric Logistic Organ Dysfunction (PELOD) score ranges from 0 to 71, with higher scores indicating more severe organ dysfunction.

¶Agents included dobutamine, dopamine (at least 5 µg per kilogram of body weight per minute), epinephrine, milrinone, norepinephrine, phenylephrine and vasopressin.

Both groups were similar in terms of types of surgical procedures, severity of illness (pediatric risk of mortality [PRISM] score, restrictive vs. liberal: 7.2 ± 4.8 vs. 7.2 ± 5.4) and proportion requiring mechanical ventilation (77% vs. 75%). An apparent difference in age between the 2 groups turned out to be statistically non significant (53.5 ± 51.8 vs. 73.7 ± 61.8 months). Table 2 displays the patient characteristics at randomization: the severity of illness (PRISM score: 3.5 ± 4 vs. 4.4 ± 4.0), frequency of MODS (35% vs. 29%) and need for mechanical ventilation (77% vs. 74%) were quite similar in the 2 groups.

Intervention

The baseline hemoglobin concentration at randomization in the restrictive and liberal groups was 7.7 ± 1.1 and 7.9 ± 1.0 g/dL, respectively (Table 2). Time between randomization and first transfusion was 0.9 ± 1.6 days versus 0.5 ± 1.7 days, ($P = 0.11$) (Table 3). As expected, hemoglobin level before the first transfusion was significantly lower in the restrictive group (6.6 ± 0.5 vs. 8.0 ± 0.9 g/dL; $P < 0.01$). After randomization, a mean difference in hemo-

globin concentration of 2.3 g/dL was observed (8.3 ± 1.1 in the restrictive and 10.6 ± 1.7 g/dL in the liberal group).

Three patients in the restrictive and 4 in the liberal group were temporarily suspended from the transfusion protocol. Such suspensions were planned in the research proposal, because participating doctors asked for the right to transfuse their patients if they became unstable or need to go to the operating room on an emergency basis. Length of suspension in the restrictive and liberal group was 1.0 ± 0.4 and 2.3 ± 2.5 days, respectively. During the suspension period, 5 red cell transfusions were given in the restrictive and 3 in the liberal group. Including the suspension period, a total of 39 transfusions were administered in the restrictive group and 109 in the liberal group ($P < 0.01$). In the restrictive group, 30 patients (50%) did not receive any red cell transfusion, whereas 62 patients (97%) in the liberal group were transfused ($P < 0.01$).

TABLE 3. Intervention (Red-Cell Transfusion), Suspension, and Cointerventions*

	Restrictive	Liberal	P
	60	64	
Red-cell transfusion and hemoglobin concentration after randomization			
Lowest Hb level (g/dL) per day in PICU	8.4 ± 1.0	10.8 ± 1.0	<0.01
Data on transfused patients			
No. transfused patients: total (%)	30 (50)	62 (97)	<0.01
Patients with 1 red-cell transfusion (%)	25 (42)	38 (59)	
Patients with 2 red-cell transfusions (%)	3 (5)	15 (23)	
Patients with more than 2 red-cell transfusions (%)	2 (3)	9 (14)	
Volume (mL/kg) per transfused patient	13.9 ± 5.6	17.6 ± 8.2	0.07
Data on first red-cell transfusion			
Days from randomization to first transfusion	0.9 ± 1.6	0.5 ± 1.7	0.11
Hb level before first transfusion (g/dL)	6.6 ± 0.5	8.0 ± 0.9	<0.01
Hb level after first transfusion (g/dL)	9.4 ± 1.3	11.0 ± 1.1	<0.01
Data on all red-cell transfusions†			
Total number of transfusions	39	109	<0.01
Volume (mL/kg) per transfusion	12.6 ± 3.7	13.6 ± 4.0	
Average length of storage (days)	16.0 ± 11.4	16.4 ± 9.2	
Longest length of storage (days)	16.7 ± 11.8	18.5 ± 9.8	
Temporary suspension from research protocol‡			
Patients suspended from study (%)	3 (5)	4 (7)	0.96
Reasons for suspension			
Acute blood loss	1	1	
Emergency surgery	2	1	
Hemofiltration	1	0	
Other	0	3	
Length of suspension (d)	1.0 ± 0.4	2.3 ± 2.5	
No. transfusion during suspension	5	3	
Cointerventions			
Fresh frozen plasma (no. patients) (%)	3 (5)	7 (10)	0.68
Platelets (no. patients) (%)	2 (3)	2 (3)	1.00
Albumin (no. patients) (%)	13 (22)	15 (23)	1.00
Corticosteroids (no. patients) (%)	9 (14)	17 (27)	0.11
Vasoactive drugs (no. patients) (%)			
Vasoactive drugs (at least 1)	21 (34)	20 (31)	
Epinephrine	13 (21)	17 (27)	
Dobutamine	12 (20)	15 (23)	
Dopamine	16 (27)	20 (31)	
Noradrenaline	12 (20)	16 (25)	

*Plus-minus values are means ± SD.

†The number is for all transfusions after randomization, including those given during suspension.

‡Attending physicians were permitted to suspend the trial and to transfuse more red cells than indicated in the study protocol if one of the following event occurred: severe acute respiratory distress syndrome with refractory hypoxemia; shock; instability in the patient's condition; acute blood loss; surgery; blood exchange-transfusion (manual or automated); hemofiltration, if priming was done with blood; or extracorporeal membrane oxygenation or plasmapheresis.

Because some data suggest that packed red blood cells with longer storage times are associated with a poorer response to transfusion,^{3,5,13} average storage time was ascertained and was 16.7 ± 11.8 days for the restrictive group versus 18.5 ± 9.8 days for the liberal group.

Cointerventions including vasoactive drugs (proportion of patients receiving at least one drug), and administration of fresh frozen plasma, platelets and albumin were similar in both groups.

Primary Outcome

Five patients in the restrictive and 6 in the liberal group developed or had worsening MODS after randomization (8% vs. 9%, $P = 0.83$) (Table 4). The absolute risk reduction was 1% (95% CI: -9% to +11%; $P = 0.83$). Since there was a nonsignificant

TABLE 4. Primary Outcome: New or Progressive MODS

	Restrictive	Liberal	ARR (%)* (95% CI)	P
No. patients (%)	60	64		
Patients with new or progressive MODS (total)†	5 (8)	6 (9)	1 (-9, +11)	0.83
Age adjusted percent (%)	(8)	(10)	2 (-8, +12)	0.75
Age				
≤28 d	1/2 (50)	0/0	NA	
29–364 d	1/12 (8)	1/14 (7)	-1 (-22, +20)	
>365 d	3/46 (7)	5/50 (10)	3 (-8, +15)	
Country				
Belgium	0/12 (0)	1/14 (7)		
Canada	5/37 (14)	4/40 (10)		
United Kingdom	0/4 (0)	0/3 (0)		
United States	0/7 (0)	1/7 (14)		
PRISM score vs. patients with primary outcome‡				
Lowest (first) PRISM = 0	0/17 (0)	0/15 (0)		
Second quartile: PRISM = 1–2	1/15 (7)	0/11 (0)		
Third quartile: PRISM = 3–5	1/16 (6)	2/15 (13)		
Highest (fourth) quartile: PRISM >5	3/12 (25)	4/23 (17)		
Suspended patients vs. with primary outcome	2/3 (67)	2/4 (50)		

*ARR: absolute risk reduction in percentage (95% CI).

†Noninferiority was checked only for the primary outcome (the number of patients who had new or progressive multiple-organ-dysfunction-syndrome (MODS), including death, after randomization). The absolute risk reduction for new or progressive MODS in the restrictive strategy group versus the liberal-strategy group was 1% (2-sided 95% CI: -9%–+11%) by intention-to-treat-analysis. Some experts also consider that a per-protocol analysis should be done in a noninferiority trial. In the per-protocol analysis, the ARR was 1% (2-sided 95% CI: -9%–+12%). The investigators and members of the Canadian Critical Care Trials Group were ready to recognize the following as a statistically significant result for this noninferiority trial: a 95% confidence interval for the absolute risk reduction that is lower than a 10% margin of safety when comparing the proportion of patients who contracted new or progressive MODS in the restrictive and liberal group. In this subgroup analysis, the 95% confidence interval was 9%, which supports the noninferiority hypothesis.

‡Pediatric Risk of Mortality (PRISM) score ranges from 0 to 76, with higher scores indicating a greater risk of death.

TABLE 5. Secondary Outcome Measures

	Restrictive	Liberal	ARR (%)* or DIM† (95% CI)	P
	60	64		
Measures of severity of organ dysfunction‡				
Highest number of organ dysfunctions	1.3 ± 1.2	1.3 ± 1.0	−0.0 (−0.4, 0.4)	
PELOD score over all PICU stay§	4.0 ± 7.1	3.5 ± 3.8	−0.5 (−2.5, 1.5)	
PELOD score on day 1§	5.3 ± 6.3	4.9 ± 5.4	−0.4 (−2.5, 0.4)	
Highest daily PELOD score after day 1§	7.4 ± 9.6	7.6 ± 8.8	0.3 (−3.0, 3.5)	
Change in PELOD score§	2.1 ± 6.3	2.8 ± 6.7	0.6 (−1.7, 2.9)	
Average daily PELOD score§	4.0 ± 7.1	3.5 ± 3.8	−0.5 (−2.5, 1.5)	
Clinical outcomes: no. patients (%)				
No. deaths in PICU	1 (2)	0 (0)		
No. deaths 28 d post PICU	0 (0)	1 (2)		
Overall 28-d mortality	1 (2)	1 (2)		
No. nosocomial infections	14 (23)	20 (31)		0.32
Other clinical outcomes (d)¶				
Length of mechanical ventilation				
After randomization	5.5 ± 5.2	6.6 ± 5.6	1.1 (−1.2, 3.4)	
Total PICU stay	6.9 ± 5.5	7.6 ± 6.1	0.7 (−1.7, 3.0)	
Length of stay from entry to PICU discharge	7.7 ± 6.6	11.6 ± 10.2	3.9 (0.8, 8.0)	0.03

*ARR: absolute risk reduction in percentage (95% CI).

†DIM: difference in means in percentage (95% CI).

‡The comparison between the restrictive-strategy group and the liberal-strategy group is given as an absolute reduction in risk.

§Pediatric Logistic Organ Dysfunction (PELOD) score ranges from 0 to 71, with higher scores indicating more severe organ dysfunction.

¶The comparison between the restrictive-strategy group and the liberal-strategy group is given in difference of the means (95% CI).

difference in age at baseline, an age-adjusted percentage for the development of MODS was calculated for each group; the results remained similar after this adjustment.

The investigators and members of the Canadian Critical Care Trials Group were ready to recognize the following as a statistically significant result for this noninferiority trial: a 95% confidence interval for the absolute risk reduction that is lower than a 10% margin of safety when comparing the proportion of patients who contracted new or progressive MODS in the restrictive and liberal group. In this subgroup analysis, the 95% confidence interval was 9%, which supports the noninferiority hypothesis.

We performed a per-protocol analysis of the primary outcome (data not shown). The results of the per-protocol analysis (absolute risk reduction in the liberal group: 1%; 95% CI: −9 to +12; $P = 0.37$) were similar to those obtained with the intent-to-treat analysis.

Secondary Outcomes

There were no differences in any of the various measures of organ dysfunction (Table 5). There was one death during PICU stay in the restrictive group and one death within 28 days postrandomization in the liberal group. No significant differences were observed in oxygenation markers or in the duration of mechanical ventilation. There were 14 cases (23%) of nosocomial infections in the restrictive group compared with 20 (31%) in the liberal group ($P = 0.32$). There was a statistically significant difference reduction in the length of PICU stay in the restrictive group when compared with the liberal group (7.7 ± 6.6 vs. 11.6 ± 10.2 days, respectively; $P = 0.03$).

DISCUSSION

In this subgroup analysis of surgical patients enrolled in the TRIPICU trial, a restrictive red-cell transfusion strategy, as compared with a liberal one, was not associated with a significant difference in new or progressive MODS. Although there was an apparent statistically significant reduction in the PICU length of stay in the restrictive group, we prudently refrain from giving any undue

emphasis to this positive finding because of the inherent limitations of subgroup analyses and because this finding cannot be explained by a difference in disease severity between the 2 groups at randomization or during their PICU stay.

These results suggest that a restrictive transfusion threshold of 7.0 g/dL is acceptable and safe for hemodynamically stable surgical patients admitted to the PICU, and that this particular subgroup of patients experienced similar outcomes to the other participants in the TRIPICU trial.

There are several benefits associated with the reduction of the number of transfusions given to critically ill patients. In this subgroup analysis, we found that a restrictive strategy results in a 2 folds reduction in both the number of surgical patients who received a transfusion as well as the total number of red-cell transfusions. Thus, a restrictive transfusion strategy allows a significant reduction in both exposure to blood products and the risks associated with transfusion without increasing the rate of new or progressive MODS, mortality, or any other MODS descriptor.

The physiologic rationale for maintaining a higher level of hemoglobin in the critically ill patients is to improve oxygen delivery. However, many studies have failed to show that red cell transfusions for this purpose does result in a significant increase in oxygen consumption/utilization after transfusion.⁵ Furthermore, there are well-known risks associated with transfusing red blood cells, such as infectious disease transmission and transfusion-related reaction such as allergic reactions and transfusion related acute lung injury (TRALI).¹⁴ Other effects of transfusion include immune suppression and systemic inflammatory stimulation and blood flow disturbance in small vessels which may contribute to MODS.^{15–17} A meta-analysis performed by Hill et al demonstrated that blood transfusions significantly increased the rate of bacterial infection in postoperative patients.¹⁸ The suggested mechanism for this infectious risk is the immunosuppressive effects of transfusions, although the authors did not mention if the packed red blood cells were leukocyte reduced.

We reviewed the transfusion strategies advocated by surgical institutions and textbooks and found that no consensus prevails on what can be the optimal transfusion threshold. Indeed, there is a lot of variability in the recommendations made to surgeons, few of which are evidence-based. While more recent surgical literature proposes transfusion thresholds as low as 7.0 g/dL for asymptomatic patients,^{19,20} other references suggest that when hemoglobin is between 6.0 and 10.0 g/dL, the treating physician can decide whether to transfuse by assessing patient status and risk of further bleeding.²¹ Many surgeons appear to be uncomfortable with the restrictive transfusion thresholds recommended for medical patients. A recent observational study published by Basora et al²² described transfusion practices in adult surgeons in Spain and reported that the mean transfusion threshold was of 8.4 ± 1.3 g/dL, a threshold somewhat higher than that recommended by the TRICC study (Transfusion Requirement In Critical Care).¹ In the United States, the CRIT study (Anemia and blood transfusion in the critically ill—current clinical practice in the United States) conducted in 284 intensive care units in 213 US hospitals in which 4892 patients were analyzed, showed that the hemoglobin concentration observed before a transfusion in adult surgical patients was higher than in medical patients (8.8 ± 1.8 g/dL vs. 8.2 ± 1.6 g/dL).²³ The evidence for transfusion practices in specific pediatric populations is scant.

Only a few studies have been conducted to evaluate the optimal transfusion threshold in surgical patients and only one of them was a clinical trial.²⁴ In the latter study, patients having undergone surgical repair for a hip fracture were randomly assigned to a symptomatic transfusion strategy or to a threshold transfusion strategy (Hb >10.0 g/dL). There was a significant difference in the median number of units transfused (2 vs. 0) as well as the mean hemoglobin levels (1.0 g/dL) in the 2 groups. The trial did not provide any conclusive evidence regarding morbidity and mortality. Thus, objective evidence to guide transfusion practices in surgical patients is scarce, especially for pediatric patients.

This is the first evidence prospective trial evidence to demonstrate that a restrictive transfusion threshold in critically ill pediatric surgical patients is safe. This trial has several other strengths. We recruited children from 17 different PICUs globally, and thus the patients are representative of the surgical cases admitted to PICU, and the results generalizable. The trial was pragmatic, and by allowing suspension from the study protocol for unstable patients, is reflective of real life clinical situations of the critically ill surgical patient. Finally, adherence to the research protocol was excellent and no patients were lost to follow up.

There are potential limitations to this trial, the most important one being that this is a subgroup analysis. Thus, it is only possible to generate hypothesis and we cannot make any definitive conclusion.^{25,26}

There is also the possibility of a site-related patient bias. Only centers whose general surgeons and intensivists were willing to accept a lower level of hemoglobin for their patients were included in the study. However, we do not believe that this bias has a significant impact because the cases and surgical procedures were varied and in sufficient numbers to adequately represent the majority of PICU surgical patients. A lack of representativeness is also possible because some children may have received transfusions during or immediately after surgery. In these patients, more elevated hemoglobin to levels above 9.5 g/dL would have precluded inclusion into the TRIPICU study because their hemoglobin level may never have dropped to the level which would have triggered recruitment. We have no evidence that this happened, but the transfusions administered prior to PICU admission were not compiled in the TRIPICU study database.

The goal of a noninferiority analysis is to establish whether the strategy of interest leads to outcomes as good as another more conventional strategy. To prove that a strategy is superior to another, a superiority analysis is required. Based on our data, the sample size required to perform such a superiority RCT would be 280 patients per arm if a 2-sided *P* value of 0.05 and a power of 0.90 are used and 215 patients per arm if the power is decreased to 0.80.

While we are aware of the limitations inherent to any subgroup analysis, several considerations prompt us to think that a larger trial is not mandatory. First, although the sample size is too small to get definitive statistical results, the data from this subgroup analysis are strikingly similar to those of the original TRIPICU study with regards to the primary outcome. There were no differences in any of the cointerventions. Almost all secondary outcomes (28-day mortality, number of nosocomial infections, length of mechanical ventilation, average daily PELOD score) were similar. Thus, it seems reasonable to conclude that a restrictive strategy for PICU surgical patients is probably safe, and that it certainly allows a reduction in the number of transfusion without changing the outcome.

CONCLUSIONS

We agree with Soliman and Cassara²⁷ who suggested that: "The decision to transfuse children should not be guided solely by the patient's hemoglobin." However, hemoglobin concentration remains the most important parameter on which practitioners base their decision to prescribe a RBC transfusion.² In this analysis of the general surgery subgroup from the TRIPICU study, we found no evidence of harm in adopting a restrictive transfusion strategy when prestorage leukocyte-reduced red cell units are used. We therefore recommend a transfusion threshold of 7.0 g/dL for stable PICU surgical patients to reduce exposure to blood products and the associated complications. This recommendation is supported by The British Society of Hematology, who suggests the acceptance of a postoperative hemoglobin level of 7.0 g/dL in both children and adults when there is good postoperative cardiac function.²⁸

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